

Decentralized Industrial Hazard Evacuation System

Complete Physical Implementation, Model Deployment & Building Installation Guide

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Target Hardware: ESP32-S3 / JetsonLife Safety Codes: NFPA 101, OSHA

Classification: Field Engineering

Orin Nano

1910, NBC Part 4

Standard

Document Purpose & Scope

This comprehensive manual provides electrical engineers, industrial facility managers, and commissioning technicians with the exact step-by-step procedures required to physically build, configure, and install autonomous **Decentralized ST-TBA-GAT Edge Signboard Nodes** into petrochemical complexes, refineries, chemical plants, and high-occupancy commercial structures.

1. PHYSICAL HARDWARE ARCHITECTURE & COMPONENT BILL OF MATERIALS

The system completely replaces vulnerable centralized SCADA wiring with autonomous, interconnected edge nodes. Each installation point operates on a modular 3-tier architecture built inside an industrial flame-retardant enclosure:

Subsystem	Physical Hardware Specifications	Electrical Interface & Industrial Standards
Edge AI Router Unit	ESP32-S3-WROOM-1U (Dual-Core Xtensa LX7 @ 240MHz, 8MB PSRAM, 16MB Flash) OR NVIDIA Jetson Orin Nano (for camera visual crowd tracking nodes).	Direct 3.3V logic, hardware cryptographic accelerator (AES-128/256), external 2.4 GHz SMA dipole antenna (+5 dBi).
Dynamic Directional Signboard	High-Luminance RGB LED Matrix (WS2812B IP67 sealed strip or HUB75 64×32 matrix, 4500 nits daylight visible). Displays: Green Arrow, Caution Amber, Red [X] Barrier, or Static White/Green Exit.	5V DC @ 4A peak, optoisolated GPIO 18 (WS2812B DIN) with 470Ω buffer resistor and 1000μF bypass capacitor. Photocell for auto-dimming.
Multi-Spectral Sensor Pod	Gas Sensor: MQ-2 / MQ-7 electrochemical (0–1000 ppm CO, flammable hydrocarbons, H₂S, Cl₂). Temperature/Humidity: Sensirion SHT31-DIS (±0.2°C, 0.1s response). Presence/Occupancy: ST VL53L1X Time-of-Flight IR Lidar (4m range, 50Hz).	- Gas: Analog ADC Pin 34 via 12-bit SAR ADC. - Temp: I2C (SDA Pin 21, SCL Pin 22) @ 400 kHz. - IR ToF: I2C bus sharing with hardware interrupt on Pin 5.
Life-Safety Actuator & Audio	Acoustic Strobe: MAX98357A I2S Class D Audio Amplifier + 85 dB directional siren. Fire Door Interlock: Solid State Relay (SSR) 5V to 24V DC solenoid lock release.	- I2S audio: BCLK Pin 26, LRC Pin 25, DIN Pin 27. - Relay output: Active-LOW GPIO Pin 23 with flyback diode (NFPA 72 fail-safe open).
Industrial Power Supply & Battery	- Primary: 24V DC Industrial Bus or PoE+ (IEEE 802.3at). - Secondary Backup: 3.2V 6000mAh LiFePO4 battery pack with integrated BMS.	Provides > 4.5 hours autonomous operation during complete facility blackouts (exceeds NFPA 101 90-minute emergency requirement).

PHYSICAL HARDWARE INTERCONNECTION SCHEMATIC:

```
+-----+
| [24V DC Main Bus / PoE+] ---> [Buck Step-Down 5V/3.3V] <---> [LiFePO4 4-Hr Battery BMS] |
| | |
| +-----v-----+ |
| [I2C: SHT31 Temp / VL53L1X ToF] -> | |--> [GPIO 18] -> [HUB75/WS2812 Sign] |
| [ADC: MQ-2 Toxic Gas Sensor] ---->| ESP32-S3 |--> [GPIO 23] -> [SSR Door Lock] |
| [RF: 2.4GHz IEEE 802.15.4 Mesh] <->| Edge AI Node |--> [I2S Bus] -> [85dB Sounder] |
| +-----+ |
+-----+
```

Reflexive Safety Hardware Interlock (Zero-Latency Override)

The microcontroller firmware implements a non-maskable hardware interrupt loop running at 100 Hz. If local toxic gas crosses 50 ppm or temperature exceeds 60°C, the firmware bypasses neural inference entirely and asserts a hardwired RED [X] barrier in **< 20 milliseconds**, preventing personnel from stepping into flashovers even during complete radio mesh loss.

2. POST-TRAINING MODEL EXPORT, OPTIMIZATION & EDGE CONVERSION

Once the ST-TBA-GAT reinforcement learning policy is trained (e.g. on GPU/Kaggle) and saved as `checkpoints/best_policy.pt`, it must be compiled, quantized, and packaged for microcontrollers.

Step 2.1: Model Export to Standard ONNX

The policy consists of the Spatio-Temporal Graph Attention network coupled to a Recurrent GRU memory unit. We export the Actor component using standard torch ONNX tracing:

```
# Command executed in terminal:
python -c '
import torch
from simulator.policy.st_tba_gat import STTBAGATPolicy

# 1. Instantiate policy and load best trained checkpoint
model = STTBAGATPolicy(node_feat_dim=34, hidden_dim=64, num_actions=7)
checkpoint = torch.load("checkpoints/best_policy.pt", map_location="cpu")
model.load_state_dict(checkpoint["model_state_dict"])
model.eval()

# 2. Trace dummy inputs for local node inference
dummy_feat = torch.randn(1, 34)          # Local 34-feature sensor observation
dummy_edge = torch.zeros((2, 6), dtype=torch.long) # Local corridor adjacency
dummy_hidden = torch.zeros(1, 64)        # Prior GRU recurrent hidden state

# 3. Export to optimized ONNX graph
torch.onnx.export(
    model,
    (dummy_feat, dummy_edge, dummy_hidden),
    "models/policy.onnx",
    input_names=["node_features", "edge_index", "hidden_state"],
    output_names=["action_logits", "new_hidden_state", "value"],
    opset_version=14,
    dynamic_axes={"edge_index": {1: "num_edges"}}
)
print("Successfully exported policy.onnx (73.0 KB)")
'
```

Step 2.2: Post-Training Quantization (FP32 to INT8)

To ensure lightning-fast deterministic inference on low-power edge chips without hardware floating-point units:

- **Quantization Tool:** ONNX Runtime INT8 Static Quantizer or TensorFlow Lite Micro Converter.
- **Calibration Dataset:** 200 recorded cycles of simulated industrial disaster telemetry (fire, toxic gas, crowd congestion).
- **Size Reduction:** 480 KB (FP32) → **73 KB (INT8)** (84.8% memory reduction).
- **Edge Latency:** < 12 ms execution time per step on ESP32-S3 @ 240 MHz (well within the 2.5-second decision interval).

Step 2.3: Compiling Model into Embedded C++ Firmware

For pure microcontroller deployment without an operating system, the INT8 weights are converted into an immutable C byte array header file:

```
# Convert quantized flatbuffer or ONNX binary to C++ header:
xxd -i models/policy_quant.tflite > embedded/observer_node/lib/hazard_core/policy_weights.h
```

3. NODE PROVISIONING & CONFIGURATION (YAML SETUP)

Every physical node installed in the facility must receive a unique configuration file matching the blueprint topology. The file defines its spatial coordinates, floor, physical neighbor corridors, doorway widths, and the precomputed NFPA static exit vector.

Step 3.1: Defining `node_config.yaml` for Physical Signboards

Example configuration for an edge signboard installed at the `hazmat_basin` intersection (Floor 1):

```
node:
  id: "hazmat_basin"           # Unique topological node ID matching blueprint
  type: "CORRIDOR_INTERSECTION" # INTERSECTION, ROOM, EXIT, or STAIRWELL
  floor: 1                     # Floor level (1, 2, or 3)
  position_meters: [45.2, 38.0, 0.0] # Precise CAD plant coordinates (X, Y, Z)
  capacity_people: 25          # Maximum NFPA corridor occupant capacity before crowd jamming
  doorway_width_m: 2.4         # Physical corridor egress width in meters

static_egress:
  default_exit_target: "stair_north_f1" # Precomputed NFPA static shortest exit corridor
  static_exit_distance_m: 67.0          # Topological distance to nearest perimeter muster point

neighbors:
  # Ordered list of up to 6 physical incident corridors
  corridor_0: { id: "tank_farm_a", distance_m: 25.0, width_m: 2.0, is_dead_end: true }
  corridor_1: { id: "tank_farm_b", distance_m: 25.0, width_m: 2.0, is_dead_end: false }
  corridor_2: { id: "pipe_rack_junc_1", distance_m: 22.0, width_m: 2.4, is_dead_end: false }
  corridor_3: { id: "stair_north_f1", distance_m: 30.0, width_m: 1.8, is_stair: true }

hardware_io:
  display_type: "HUB75_64x32" # Dynamic RGB LED directional matrix
  audio_i2s_pin: 25           # Voice strobe alarm pin
  sensors:
    gas_mq_analog_pin: 34      # Toxic gas analog input
    temp_sht31_i2c_addr: 0x44  # Ambient thermal sensor
    ir_presence_interrupt_pin: 5 # Optical crowd flow counter

mesh_network:
  rf_protocol: "IEEE_802.15.4_ESP_NOW" # 2.4 GHz ad-hoc peer mesh
  channel: 11                          # Protected non-overlapping industrial Wi-Fi/ZigBee channel
  pan_id: "0x7B9A"                     # Facility plant mesh PAN ID
  aes_gcm_key: "8F4A12B9D8E7340156C9A20F88319E4D" # 128-bit AES pre-shared mesh encryption key
```

Step 3.2: Firmware Flashing via PlatformIO CLI

Connect the ESP32-S3 edge node via USB-C to the engineering laptop and execute:

```
# 1. Navigate to embedded node directory
cd embedded/observer_node

# 2. Compile and upload firmware directly to ESP32-S3
pio run -t upload -e esp32-s3-devkitc-1

# 3. Write specific node configuration to onboard SPIFFS / LittleFS partition
pio run -t uploadfs

# 4. Open serial monitor to verify sensor zero-baseline and RF mesh pairing
pio device monitor -b 115200
```

4. PHYSICAL BUILDING INSTALLATION & MOUNTING GUIDE

To ensure full compliance with international life-safety standards (NFPA 101 Life Safety Code, OSHA 1910.37, and Indian National Building Code Part 4), physical mounting and wiring must strictly adhere to the following architectural guidelines:

4.1 Signboard Placement & Visual Sightlines

Overhead High-Level Signboards:

- **Mounting Height:** Bottom edge must be installed between **2.0 meters and 2.5 meters (6.5 to 8.2 ft)** above finished floor level (AFFL).
- **Location:** Installed at every corridor decision point, T-intersection, stairwell entrance, and ramp departure.
- **Viewing Distance:** Maximum continuous corridor spacing must not exceed **30 meters (100 ft)** to guarantee 100% uninterrupted visual line-of-sight through smoke.

Low-Level Photoluminescent Floor Beacons:

- **Mounting Height:** **150 mm to 200 mm (6 to 8 inches)** above the floor surface.
- **Critical Purpose:** In severe thermal fires, hot toxic smoke stratifies near the ceiling. Workers are trained to crawl; low-level dynamic LED arrows provide life-saving visibility under the dense smoke layer.

4.2 Sensor Pod Placement Criteria

- **Thermal Sensors:** Positioned in the upper third of the wall, exactly **0.3 meters below the ceiling slab** to capture rising convective heat plumes early.
- **Gas Sensors:**
 - For *heavier-than-air* gases (H₂S, Chlorine, Butane): Install auxiliary sensor pod at **0.5 meters AFFL** near floor trenches.
 - For *lighter-than-air* gases (Methane, Hydrogen, Ammonia): Install at ceiling height.
- **Optical Crowd Sensors:** Mount directly above corridor choke points and doorways pointing downward at a 45-degree angle to count passing heads without occlusion.

4.3 Electrical Wiring & Cable Routing

SAFETY MANDATE: Fire-Rated Cabling & Conduit

All primary 24V DC power feeds must be enclosed in galvanized rigid steel (GRS) electrical conduit or mineral-insulated copper-clad (MICC) **2-hour fire-rated cabling (CW1733 / BS 6387 CWZ)**. In hazardous production areas (ATEX Zone 1/2), use explosion-proof junction boxes with silicone gaskets.

CORRIDOR ELEVATION & SIGNBOARD INSTALLATION PROFILE:

Ceiling Slab [H = 3.6m] -----+
| [Temp / Smoke Sensor Pod: 0.3m below ceiling] |
| |
| +-----+ |
| | [DYNAMIC LED SIGNBOARD: GREEN ARROW / CAUTION / RED] | |
| +-----+ |
| Mounting Height: 2.2m AFFL (Clear Headroom Clearance) |
| |
| |
| [Low-Level Smoke Floor Arrow: 0.2m AFFL] |
Finished Floor [0.0m] =====+

5. COMMISSIONING, CALIBRATION & FIELD ACCEPTANCE TESTING

Before declaring any industrial facility zone active, the commissioning engineer must execute and sign off on the 5-stage Field Verification Protocol:

Stage 1: RF Mesh Link Quality Verification (Site Survey)

- Power on all edge signboards across the floor.
- Using an RF spectrum analyzer or the ESP32 serial console (`mesh_diag`), measure Received Signal Strength Indicator (RSSI) between every 1-hop neighbor:
 - **Acceptable Range:** RSSI ≥ -75 dBm (Packet Error Rate < 1%).
 - **Marginal Warning:** -75 dBm > RSSI ≥ -85 dBm (Install RF mesh repeater node).
 - **Unacceptable:** RSSI < -85 dBm (Corridor obstruction requires repositioning external antenna).

Stage 2: Sensor Baseline & Zero-Drift Calibration

- Ensure plant ventilation is running with clean ambient air.
- Trigger zero-point calibration routine via mesh broadcast: `broadcast_cmd --calib-zero-all` .
- Verify that baseline readings settle at: Temperature: 20 - 25°C, Gas: 0.0 ppm, Presence: 0 count.

Stage 3: Signboard Actuator Display Test

Execute visual command test sequence on every signboard:

1. **Code 0 (Standby / Normal):** Signboard displays standard static exit vector in crisp white/green illuminated text.
2. **Code 1 (Caution / Detour):** Signboard illuminates amber chevron arrows pulsing at 1 Hz.
3. **Code 2 (BLOCKED / Red X):** Signboard displays high-intensity Red [X] barrier; relay triggers fire door closure.
4. **Code 3 (Dynamic Green Arrow):** Signboard lights green chase arrows pointing along the selected exit route.

Stage 4: Hardware Reflexive Safety Interlock Validation

- Apply calibrated test aerosol (CO test spray or heat gun @ 65°C) directly to the sensor pod.
- **Requirement:** Verify with a high-speed digital timer that the local signboard flips to **RED [X]** in < 20 milliseconds, and adjacent nodes receive the 4-byte delta gossip packet within < 50 ms.

Stage 5: Evacuation Drill & Egress Throughput Validation

- Simulate a catastrophic incident at `reactor_1` via engineering console.
- Verify that 100% of signboards dynamically reconfigure:
 - Nodes adjacent to `reactor_1` lock down with Red [X].
 - Mid-plant corridors split crowd flow between `muster_point_alpha` and `muster_point_bravo` .
 - Zero evacuees enter dead ends or loops.

Commissioning Sign-Off Checklist

- ☐ All 36 node IDs match CAD blueprint plant coordinates.
- ☐ RF Mesh packet delivery rate exceeds 99.2% across multi-floor stairwells.
- ☐ LiFePO4 battery discharge test verified > 4.5 hours continuous illumination.
- ☐ Hardware fail-safe override confirmed under physical aerosol challenge.
- ☐ Safety Authority (NFPA Inspector / Factory Inspectorate) approval signed.

6. ROUTINE MAINTENANCE & INSPECTION SCHEDULE

Frequency	Inspection & Maintenance Tasks	Pass / Fail Criteria
Monthly	Automated mesh self-diagnostic test; LED pixel health test; optical sensor dust wipe.	0 dead pixels; 100% mesh nodes reporting heartbeat at 0.1 Hz.
Quarterly	Physical bump-test of toxic gas sensors using test gas canister; manual fire door release test.	Sensor response time < 3.0s; door release time < 1.0s.
Annually	Full battery capacity discharge test under maximum LED load (simulated power blackout).	Battery voltage must remain above 2.8V per cell after 90 minutes.